24ADC001 — BUSINESS INTELLIGENCE FOR DECISION MAKING

LAB: THE INVESTMENT COMMITTEE PITCH

A risk-vs-return lab with a single quantitative outcome: your portfolio's Sharpe Ratio

The Scenario

You are a junior analyst at a mid-sized investment firm. The firm is considering five funds for its next client offering, and the Investment Committee meets Friday. Your job: analyze all five funds, then build and defend a three-fund portfolio. The Committee doesn't want opinions — they want a number. Your portfolio will be judged on one figure: its Sharpe Ratio, the industry-standard measure of return earned per unit of risk taken.

Concepts covered: Expected Return, Standard Deviation, and the Sharpe Ratio (risk vs. return analysis).

The Data

File: fund_returns.csv — five funds' annual returns over the last five years.

Fund	Year 1	Year 2	Year 3	Year 4	Year 5
GreenTech Ventures	18%	−6%	32%	22%	14%
Steady State Bonds	6%	5%	7%	6%	8%
Metro Realty Trust	10%	14%	−3%	19%	12%
Consumer Staples Fund	9%	11%	8%	10%	9%
Frontier Markets Fund	25%	−15%	40%	5%	30%

Assume a risk-free rate of 4% throughout.

Part 1: Analyze Each Fund Individually

1. For each of the five funds, compute the Expected Return: $E(R) = \Sigma(\text{Returns}) / n$
2. For each fund, compute the Standard Deviation (risk): $\sigma = \sqrt{[\Sigma(R - E(R))^2 / n]}$
3. For each fund, compute the Sharpe Ratio: $\text{Sharpe} = (E(R) - \text{Risk-Free Rate}) / \sigma$
4. Rank all five funds by Sharpe Ratio, highest to lowest. Note which fund has the highest raw expected return — is it the same fund that tops your Sharpe Ratio ranking?

Part 2: Build a Three-Fund Portfolio

Choose exactly three of the five funds and decide what percentage of the budget to put in each, subject to:

- All three weights must sum to exactly 100%
- No single fund may receive less than 10% or more than 70%

Compute your portfolio's combined figures using a simplified weighted-average method (the same simplification used in this unit's Sharpe Ratio material — a full treatment would also account for how the funds move together, which is beyond this lab):

- Portfolio $E(R) = \Sigma(\text{weight}_i \times E(R)_i)$
- Portfolio $\sigma = \Sigma(\text{weight}_i \times \sigma_i)$
- Portfolio Sharpe Ratio = (Portfolio E(R) − 4%) / Portfolio σ

This single number — your portfolio's Sharpe Ratio — is your quantitative outcome for this lab.

Part 3: Defend It to the Committee

Write a 200-word pitch (as if speaking to the Investment Committee) covering:

- Which three funds you chose and your exact allocation
- Your portfolio's E(R), σ, and Sharpe Ratio
- Why you did NOT include the two funds you left out — name them specifically and explain what made them less attractive for this portfolio
- One risk you are knowingly accepting with your chosen allocation

Deliverables

- A table with all five funds' E(R), σ, and Sharpe Ratio, and your ranking
- Your chosen three funds, weights, and portfolio E(R) / σ / Sharpe Ratio, with the arithmetic shown
- The 200-word pitch

Grading Rubric (20 marks)

Criterion	What earns full credit	Marks
Individual fund calculations	All five funds' E(R), σ, and Sharpe Ratio computed correctly, with the ranking correctly identifying that the highest-return fund is not necessarily the highest-Sharpe fund	6
Portfolio construction validity	Exactly three funds chosen, weights sum to 100%, and the 10–70% constraint is respected	3
Portfolio Sharpe Ratio achieved	Scored on a curve against the class: top-third of submitted portfolio Sharpe Ratios earn full marks, middle-third earn 6/8, bottom-third earn 4/8	8
Pitch quality	Specifically explains why the two excluded funds were left out (not just “lower Sharpe”) and names a genuine, specific risk being accepted	3